
API Endpoint Tables

AI-Powered Dental Diagnostic & Treatment System

1. Authentication APIs (/api/auth)

Handles user registration, login, profile retrieval, and account management. All protected endpoints require a valid JWT token in the Authorization header.

Method	Endpoint	Description	Request Body / Params	Response
POST	/api/auth/register	Register a new user account.	fullName, email, password, phone, profileImage	Created user object with JWT token
POST	/api/auth/login	Authenticate an existing user.	email, password	JWT token and user profile summary
GET	/api/auth/profile	Retrieve logged-in user profile.	Authorization token in header	Authenticated user details
PUT	/api/auth/profile	Update user profile information.	fullName, phone, profileImage	Updated user object
DELETE	/api/auth/profile	Deactivate or delete user account.	Authorization token in header	Success message / deactivation confirmation

2. Medical Record APIs (/api/records)

Manages the Comprehensive Medical Resource & History Vault. Supports full CRUD for patient medical records including X-rays, photos, clinical reports, schedules, and inventory. Soft-delete is enforced — records are never physically removed. The `file_url` field bridges this module with the AI/ML diagnostic engine.

Method	Endpoint	Description	Request Body / Params	Response
POST	<code>/api/records</code>	Create a new patient medical record.	<code>patient_id</code> , <code>doctor_id</code> , <code>record_type</code> , <code>title</code> , <code>description</code> , <code>record_date</code> , <code>file_url</code>	201 — Created record object
GET	<code>/api/records/patient/:patient_id</code>	Get full history for a patient (non-deleted).	Path param: <code>patient_id</code>	200 — Array of record objects sorted by date desc
PUT	<code>/api/records/:record_id</code>	Update allowed fields of a record.	<code>title</code> , <code>description</code> , <code>record_type</code> , <code>file_url</code> , <code>record_date</code>	200 — Updated record object
DELETE	<code>/api/records/:record_id</code>	Soft-delete a record (Admin only).	Header: <code>x-user-role: admin</code> ; Path param: <code>record_id</code>	200 — Success message / 403 Forbidden

3. AI Diagnostic APIs (`/api/diagnostic`)

Powers the Multi-Condition Dental Diagnostic Engine (MC-DDE). Accepts dental photographs or X-rays and returns simultaneous detection results for four conditions: caries, malocclusion, oral wounds, and calculus. Inference results are stored and linked to patient records via the Medical Vault.

Method	Endpoint	Description	Request Body / Params	Response
POST	/api/diagnostic/analyze	Submit dental image for multi-condition AI inference.	patient_id, image (base64 or file_url), input_type (xray/photo)	Diagnostic results: caries, malocclusion, wounds, calculus with severity scores
GET	/api/diagnostic/results/:patient_id	Retrieve all past diagnostic results for a patient.	Path param: patient_id	Array of diagnostic result objects
GET	/api/diagnostic/results/detail/:id	Retrieve a single diagnostic result with visual overlay data.	Path param: id	Single diagnostic result with overlay metadata
POST	/api/diagnostic/report/:id	Generate and save a clinical report from a diagnostic result.	Path param: id	Generated clinical report object with treatment recommendations

4. Appointment Scheduling APIs (/api/appointments)

Manages the booking of doctor consultations. Prevents double bookings by validating time-slot availability before confirming. Appointments align with Medical Record dates for seamless history tracking.

Method	Endpoint	Description	Request Body / Params	Response
POST	/api/appointments	Book an appointment with a doctor.	patient_id, doctor_id, appointment_date, time_slot	201 — Created appointment object
GET	/api/appointments	Retrieve all appointments (Admin).	Optional query: status, doctor_id, date	200 — List of appointment records
GET	/api/appointments/patient/:patient_id	Retrieve upcoming and past appointments for a patient.	Path param: patient_id	200 — Patient appointment history
PUT	/api/appointments/:id	Reschedule or update an appointment.	appointment_date, time_slot, status	200 — Updated appointment object
DELETE	/api/appointments/:id	Cancel or delete an appointment.	Path param: id	200 — Cancellation confirmation

5. Doctor's Digital Prescription APIs (/api/prescriptions)

Allows dentists to create, view, edit, and manage digital prescriptions. Prescriptions in draft state can be modified or deleted. Final prescriptions are linked to medication orders and patient records.

Method	Endpoint	Description	Request Body / Params	Response
POST	/api/prescriptions	Create a digital prescription for a patient.	patient_id, doctor_id, medications (array: name, dosage, frequency), notes	201 — Created prescription object
GET	/api/prescriptions/:id	Retrieve a single prescription by ID.	Path param: id	200 — Single prescription details
GET	/api/prescriptions/patient/:patient_id	Retrieve all prescriptions for a patient.	Path param: patient_id	200 — Array of prescription objects
PUT	/api/prescriptions/:id	Modify a prescription before final submission.	medications, notes	200 — Updated prescription object
DELETE	/api/prescriptions/:id	Delete a prescription draft (Doctor only).	Path param: id	200 — Deletion confirmation

Note: Only the prescribing doctor can delete a draft prescription. Once a medication order is created from a prescription, the prescription is locked from further edits.

6. Medication Ordering & E-Payment APIs (/api/orders)

Enables patients to order medications directly from approved prescriptions. Card details are validated for correct format only — no real payment gateway is integrated in this academic phase. Admin verification is required before an order is confirmed.

Method	Endpoint	Description	Request Body / Params	Response
POST	/api/orders	Create a medication order from a prescription.	patient_id, prescription_id, medications, total_amount, payment_details	201 — Created order object with status: Pending
GET	/api/orders	Retrieve all orders (Admin).	Optional query: status, patient_id	200 — List of order records
GET	/api/orders/patient/:patient_id	Retrieve order history for a patient.	Path param: patient_id	200 — Patient order history
GET	/api/orders/:id	Retrieve a single order by ID.	Path param: id	200 — Single order details
PUT	/api/orders/:id	Update order status after payment confirmation.	status, payment_status, payment_date	200 — Updated order object
DELETE	/api/orders/:id	Delete an unpaid/pending medication order.	Path param: id	200 — Deletion confirmation

7. Feedback & Rating APIs (/api/feedback)

Allows patients to submit a star rating and written comment after a completed consultation. Feedback is editable within an allowed time window. Doctors can view their aggregate rating summary through this API.

Method	Endpoint	Description	Request Body / Params	Response
POST	/api/feedback	Submit a rating and feedback for a consultation.	patient_id, doctor_id, appointment_id, rating (1–5), comment	201 — Created feedback object
GET	/api/feedback/doctor/:doctor_id	Retrieve all feedback for a doctor.	Path param: doctor_id	200 — Array of feedback with aggregate rating
GET	/api/feedback/patient/:patient_id	Retrieve all feedback submitted by a patient.	Path param: patient_id	200 — Patient feedback history
PUT	/api/feedback/:id	Edit feedback within the allowed time period.	rating, comment	200 — Updated feedback object
DELETE	/api/feedback/:id	Delete a feedback entry.	Path param: id	200 — Deletion confirmation

8. Emergency Contact Management APIs (/api/emergency-contacts)

Allows patients to register and manage emergency contact details. A primary contact flag can be set. All contacts are linked to the patient's ID for quick retrieval during emergency scenarios.

Method	Endpoint	Description	Request Body / Params	Response
POST	/api/emergency-contacts	Add a new emergency contact for a patient.	patient_id, name, relationship, phone, is_primary	201 — Created contact object
GET	/api/emergency-contacts/:patient_id	Retrieve all emergency contacts for a patient.	Path param: patient_id	200 — Array of contact objects
PUT	/api/emergency-contacts/:id	Update emergency contact information or priority.	name, relationship, phone, is_primary	200 — Updated contact object
DELETE	/api/emergency-contacts/:id	Remove an emergency contact from the system.	Path param: id	200 — Deletion confirmation

9. Image Upload APIs (/api/upload)

Handles image uploads for dental diagnostic images (X-rays, intraoral photos), facility records, and user profile pictures. Uploaded file URLs are stored and referenced in the relevant record fields including file_url in Medical Records and profileImage in User Auth.

Method	Endpoint	Description	Request Body / Params	Response
POST	/api/upload	Upload dental images, X-rays, or facility photos.	multipart/form-data: image file, category (record/facility/profile)	Uploaded file URL / image path stored in server
DELETE	/api/upload/:filename	Remove an uploaded image if no longer needed.	Path param: filename	200 — Deletion confirmation